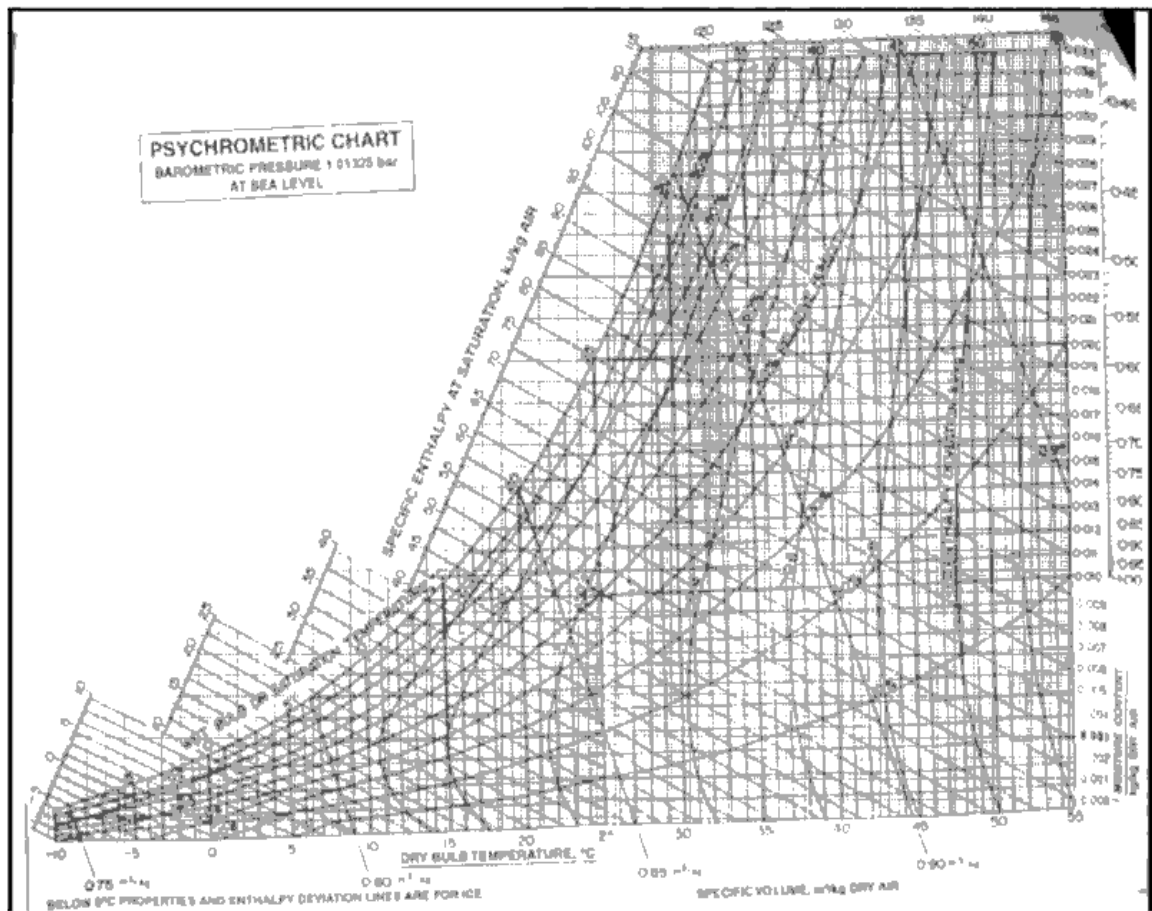


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE							
Supplementary Winter Exam-2023							
Course: B. Tech.		Branch: Mechanical & allied		Semester: VI			
Subject Code & Name: Applied Thermodynamics- II (BTME603_Y19)							
Max Marks: 60		Date: 20.01.2024		Duration: 2:00 pm to 05:00 pm			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly. 5. Use of steam tables, Refrigerant property tables, Psychrometric charts and non –non-programmable calculators is allowed							
				(Level/CO)	Marks		
Q. 1	Solve Any Two of the following.				12		
A)	Draw a valve timing diagram for a typical 4-stroke Petrol engine. Explain, what is valve overlap.				Understand/ CO1	6	
B)	Compare 4-stroke and 2-stroke cycle engines.				Understand/ CO1	6	
C)	Oil engine working on the Diesel cycle has a compression ratio of 14. The fuel cut-off takes place at 6 % of stroke. Calculate the cut-off ratio and diesel cycle efficiency. State the assumptions made, if any. $r_c=1.78$, $\eta_{TH}=60.43\%$				Apply /CO1	6	
Q.2	Solve Any Two of the following.					12	
A)	Draw the schematic diagram Diesel engine Power Plant and State various applications of the Diesel Engine Power Plant				Understand/ CO5	6	
B)	Explain with a neat labelled diagram of a typical water-cooled system used in a four-cylinder engine.				Apply/ CO2	6	
C)	Explain the catalytic converter used in IC engine. Discuss various types of alternative fuels used for SI and CI engines in India.				Understand/ CO2	6	
Q. 3	Solve Any Two of the following.					12	
A)	What is the difference between vapour absorption and vapour compression systems?				Apply/ CO3	6	
B)	A vapour compression refrigerator uses R-40 and operates between temp. limits of -10 °C and 45 °C. At the entry to the compressor, the refrigerant is dry saturated and after compression, it acquires a temperature of 60 °C. Find The C.O.P. of a refrigerator. The relevant properties of R-40 are as follows				Apply/ CO3	6	
		Saturation Temp. in °C		Enthalpy in KJ/Kg		Entropy in KJ/Kg K	
				Liquid	Vapour	Liquid	Vapour
		-10		45.4	460.7	0.183	1.637
		45		133	483.6	0.485	1.587

C)	Discuss the desirable properties of refrigerants used in vapour compression refrigeration systems.	Understand/ CO3	6
Q.4	Solve Any Two of the following.		12
A)	Assume that the outside air temperature is 32°C with a relative humidity $\phi = 60\%$. Use the psychrometric chart to determine the 1. Specific humidity 2. The enthalpy 3. The wet-bulb temperature 4. The dew-point temperature and 5. The specific volume of the dry air . 6. Indicate all the values determined on the chart.	Apply/CO4	6
B)	Explain following terms: 1) DBT 2) WBT 3) Specific humidity 4) Relative humidity	Remember/ CO4	6
C)	Explain the following source of energy for the power plant i) Fossil fuels, ii) Petroleum products, iii) Tidal	Understand/ CO5	6
Q. 5	Solve Any Two of the following.		12
A)	Write short notes on 1. Topping cycle 2. Rankine cycle	Remember/ CO5	6
B)	Explain with a schematic diagram of a typical thermal power plant.	Apply/CO6	6
C)	Draw a schematic diagram of a typical nuclear power plant. Write down the guide-lines for the selection of a site for this type of power plant.	Remember/ CO6	6
*** End ***			

The grid and the borders of the table will be hidden before final printing.



[Note:- Use the above chart for Q.No. 4 (A)]